



Department of Electrical and Electronics Engineering
Academic Year 2023 – 2024 (Odd Semester)

Degree, Semester & Branch: VII Semester B.E EEE

Course Code & Title: EE8010 – Power System Transients

Name of the Faculty member (s): Dr. K. Karthikeyan

Innovative Practice Description

- **Unit / Topic: Unit I / Different types of power system transients**
- **Course Outcome: CO 1**
- **Topic Learning Outcome: TLO 1**
- **Activity Chosen: Mind map**
- **Justification:**
 - ✓ The chosen topic has many ways of classifying the power system transients. Each classification is having more than two categories of transients. A Mind Map is an easier way to brainstorm thoughts without worrying about order and structure. It allows one to visually structure the learnt concepts to help with recall. Hence it matches with the chosen topic.
- **Time Allotted for the Activity: 15 minutes**
- **Details of the Implementation:**
 - ✓ The concept of mind map is explained to the students. Few examples of the mind map for common topics are shown to them.
 - ✓ The students are allowed to refer their class notes and PPT for preparing their mind map. They have given with 3-4 minutes for referring the concepts.
 - ✓ The team is formed with 3-4 students in a group. Then they are allowed to draw the mind map in discussion with their team members.
 - ✓ Finally, all the team shared their mind map with other teams.
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO5	PO8	PO10	PO12	PSO1
CO1	3	1	1	1	1	1	1	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO9	PO10
	1	1
Justification for correlation	The students are working in a group to list out the points and design a mind map. Hence it is mapped with level 1.	The students are drawing a neat mind map and trying to incorporate as much as points possible. Hence their visualization skills and presentation skills are improved. It is mapped with level 1.

- **Images / Screenshot of the practice:**



- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- ✓ The students are very happy to have such a brain storming and visualizing activity in the classroom.
 - ✓ The students are also interested in such activities in future topics.

- ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- ✓ The students are able to understand and remember the concepts easily, because the Mind Map is a highly effective way of getting information in and out of the brain.
 - ✓ The students are happy that they can reproduce this mind map in the examination which is improving their presentation skills.

- ❖ ***Challenges faced in implementation:***

- ✓ Few students are not interested to participate in the activity due to their lacking ability to visualize and present their concepts learnt.
 - ✓ The time management for the activity is an issue. I could not able to complete the activity by planned time.

References:

- ❖ Allan Greenwood, 'Electrical Transients in Power Systems', Wiley Inter Science, New York, 2nd Edition, 1991.
- ❖ <https://transientspecialists.com/blogs/blog/electrical-transients>
- ❖ https://en.wikipedia.org/wiki/Mind_map

Signature of Faculty Member

HoD/EEE



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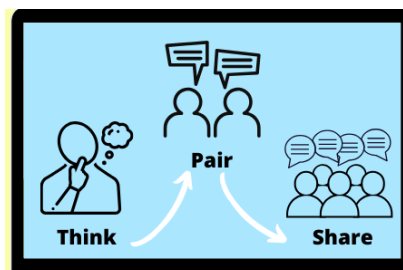
Degree, Semester & Branch: VII Semester B.E EEE

Course Code & Title: EE8010 – Power System Transients

Name of the Faculty member (s): Dr. K. Karthikeyan

Innovative Practice Description

- **Unit / Topic: Unit II / Normal and abnormal switching transients**
- **Course Outcome: CO 2**
- **Topic Learning Outcome: TLO 6**
- **Activity Chosen: Think – Pair - Share**
- **Justification:**
 - The concept of switching transients is relatively a difficult to understand by the students. Especially understanding the difference between normal and abnormal transients is quite difficult for the students. Therefore, the team activity “Think-Pair-Share” has been planned for this topic.
- **Time Allotted for the Activity: 15 minutes**
- **Details of the Implementation:**
 - The students have been grouped into smaller groups of four members, with one bright student, two average learners and one slow learner in each group. One average learner from each group has been made as a recorder.
 - I have shown the power point presentation for the quick recap of the concepts.
 - The details of the normal switching transients and the circumstances when the transients will become abnormal switching transients are explained to the students.
 - The students have been told to recollect the concepts and then asked to discuss about it within the group for 5 minutes.
 - Finally, two recorders have been asked to share their results to the whole class through chalk and talk technique. All other students interacted with them for proper understanding the concepts.



• CO – PO / PSO mapping:

CO	PO 1	PO 2	PO 3	PO 5	PO 9	PO 10	PO12	PSO 1
CO2	3	1	1	1	1	1	1	2

(1 – Low 2 – Moderate 3 – High)

• PO / PSO mapped:

Innovative practice	PO 9	PO 10
	1	1
Justification for correlation	The students will do this activity in team. Hence it is mapped with PO 9 with level 1.	The students will communicate with each other and do the oral presentation. Hence this activity is mapped with PO 10 with level 1.

• Images / Screenshot of the practice:

Team Members
P.G. Gayathree
G.Dhana Bhoothi Lakshmi

Think Pair Share

Topic : Normal and Abnormal Transients.

- * When switch is open, it is possible to reach twice as high as normal voltage
- * When switch closes, current can reach twice of steady state current.
- * Due to current damping, it is not possible in practical circuits.
- * Due to transients, the voltage and current magnitude may rise high.
- * It is also caused due to charge on a capacitor or line and current in an inductor.
- * Transients stores energy, so when second transient is initiated it would be abnormal.

TEAM MEMBERS

COURSE CODE : EE8010 UDHAYA-T- 95362010502

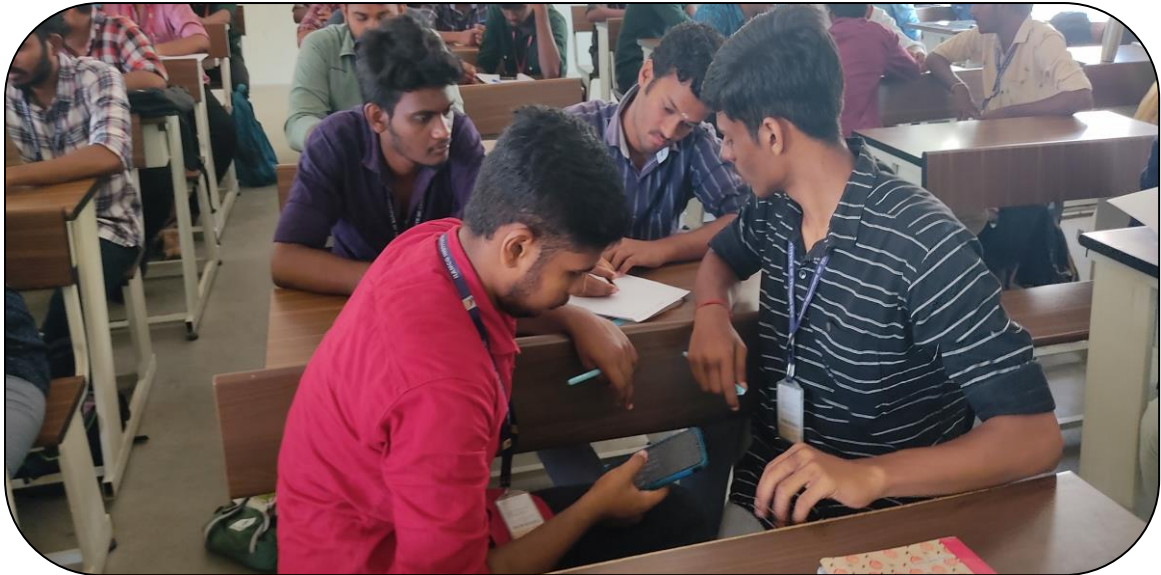
COURSE NAME : POWER SYSTEM TRANSIENTS KAVAI VIZHI-P- 953620105006

TITLE : NORMAL & ABNORMAL SWITCHING TRANSIENTS SABARISHA V- 953620105019

YUVA PRIYA-D- 953620105026

THINK , PAIR , SHARE

- * If switch opens, the recovery voltage is twice as normal peak voltage.
- * If switch closes, current twice as eventual steady state current.
- * Due to circuit damping, magnitudes of current and voltage are not achieved.
- * Due to transients, the magnitude of voltage and current rise high. Such transients are referred as abnormal current & voltage transients.
- * The transient stores energy in the system, when a second transient is initiated it would be abnormal.
- * When initial current $I_1(0)$ voltage across capacitor $V_1(0)$ and current through capacitor $I_2(0)$ is finite, then abnormal transients developed.



- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- ✓ The students have enjoyed the activity. Most of the students requested me to conduct such activities in the classes. When I did this activity, they had a great confidence in learning the new concept.

- ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- ✓ The students have understood the concepts clearly and it has been evident from the points they have shared with the classroom.
- ✓ The students have presented the points to the class; it cleared any ambiguity in understanding the important points in the chosen topic.

- ❖ ***Challenges faced in implementation:***

- ✓ The activity has been planned for 15 minutes, but it has been completed after 16 minutes only. Next time, I should plan the activity with proper time.
- ✓ The group has to form as heterogeneous, and then only I can take care about the non-participating students.

References:

- ❖ Allan Greenwood, “Electrical Transients in Power Systems”, Wiley Inter Science, New York, 2nd Edition, 1991.
- ❖ <https://powerworldacademy.com/introduction-to-power-system-switching-transients/>
- ❖ <https://www.kent.edu/ctl/think-pair-share>

Signature of Faculty Member

HoD/EEE